Probing Dimensional Crossover in SCAN Exchange with Exact Quasi-One-Dimensional Wires and Quasi-Two-Dimensional Sheets

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Abstract

Recent work by Constantin, "Dimensional crossover of the exchange-correlation energy at the semilocal level," Phys. Rev. B 78, 155106 (2008), DOI: 10.1103/PhysRevB.78.155106, motivates our dimensional-crossover extension of SCAN exchange. That work showed that a semilocal functional can recover the exact two-dimensional exchange limit if its enhancement factor decays as $s^{-1/2}$. SCAN has a single tail of this kind and cannot use it selectively: on one-orbital densities its iso-orbital indicator vanishes, so it cannot tell a sheet from a wire and misses the exact exchange limits of both. In this study, we address this failure. We have developed an exact fixed-density benchmark: two-electron anisotropic Gaussian densities from the aniGTO harmonic-trap basis, whose exact exchange is a single anisotropic repulsion integral evaluated in aniGTO. Our analysis shows that on this benchmark SCAN exchange grows as $\lambda^{1/2}$ on the wire (-292 Ha against the exact -7.90 Ha at $\lambda = 10^4$) and tends to -11.145 Ha on the sheet, 8.9 times the exact limit $-\sqrt{\pi/2}$. As a solution, we propose DC-SCAN, which adds the log-density curvature through a rank descriptor and a curvature scale, with

two coefficients fixed analytically rather than fitted. DC-SCAN removes the power-law divergence, reproduces the exact coefficient of the wire logarithm up to a constant finite-width offset of −0.040 Ha, stays within 0.10 Ha of the exact sheet exchange and reaches its limit, and leaves SCAN unchanged for isotropic and slowly varying densities. Applied to the infinite-barrier slab, our sheet construction reproduces Constantin's coefficient (0.5217) and the SCAN two-dimensional limit of Kaplan, Wagle and Perdew (−1.671 Ha) to 0.02%, and we extend the analysis to the wire limit, which has not been examined for SCAN against an exact reference. We keep the benchmark exact at extreme anisotropy with a logarithmic-variable repulsion integral, accurate to double precision where the default quadrature fails by up to 11.3 Ha, and an exact axisymmetric $N = 2$ strictly-correlated solver. These fixed-density, exchange-only results are the basis for tests on non-Gaussian profiles, three-dimensional norms, the self-consistent potential and correlation.

1 Introduction

Semilocal meta-generalized-gradient approximations (meta-GGAs) such as SCAN[1] are constructed for three-dimensional (3D) densities. In confined systems — quantum wells, wires, monolayers and anisotropic traps — the density can be compressed along one or two axes, and the exchange energy then follows a different asymptotic law: it stays finite when one coordinate is compressed and diverges logarithmically when two are, because the electrons still interact through the 3D Coulomb potential. Constantin showed that a semilocal functional recovers the exact two-dimensional (2D) exchange of an infinite-barrier slab if its enhancement factor tends to $0.5217\, s^{-1/2}$,[2] and Kaplan, Wagle and Perdew found that SCAN, alone among LDA, PBE and SCAN, reaches a finite 2D limit for this slab, but one about an order of magnitude too negative.[3] Semilocal exchange functionals collapsing from three to two dimensions have been constructed,[4] and density-Hessian (Hessian-level) meta-GGAs have been implemented self-consistently.[5] In this study, we address a narrower question: does

SCAN exchange obey the two confinement limits on solvable densities, and what is the lowest additional density-derivative information needed to make it do so?

As test densities, we use two-electron densities generated by one doubly occupied anisotropic Gaussian. They are the simplest objects of the aniGTO anisotropic Gaussian integral package:[6] the orbital is the s function of the aniGTO harmonic-trap basis, and the exact exchange energy is one anisotropic two-electron repulsion integral. Because the density is one-orbital, the SCAN iso-orbital indicator is $\alpha = 0$ everywhere for every confinement strength, so α carries no geometric information: a sphere, a sheet and a wire look the same to SCAN. Whether SCAN nevertheless respects the sheet and wire limits has not been checked against exact references. We show that SCAN then has the wrong asymptotic power on the wire and the wrong finite limit on the sheet. We then propose and test a dimensional-crossover extension of SCAN exchange (DC-SCAN) that uses the log-density curvature to select a sheet branch, a wire branch or unmodified SCAN. Our sheet construction reproduces both slab results of refs 2,3 independently, and we extend the analysis to the two-axis (wire) limit, where, to our knowledge, the behavior of SCAN has not been examined against an exact reference. We collect all definitions and derivations in Appendix A, and we give the numerical checks in the Supporting Information (SI).

2 Theory

The reference densities are the Gaussian sheet and wire families of eqs A7 and A8, with confinement parameter $\lambda \geq 1$; $\lambda = 1$ is the isotropic Gaussian. Their exact exchange is $E_x = -E_H/2 = -(\phi\phi|\phi\phi)$ (eq A9), which has the closed forms eqs A10 and A11. For strong confinement the sheet tends to the finite value $-\sqrt{\pi/2}$ (eq A12) and the wire diverges as $-\sqrt{2/\pi}\,\ln(2\lambda)$ (eq A13); a finite strict-1D exchange energy is therefore not the correct target for a wire with the 3D interaction.

On the $\alpha = 0$ branch the SCAN enhancement factor (eq A6) decays as $s^{-1/2}$. With the

scaling of the reduced gradient on the two families (eq A14) this gives a finite sheet energy but $E_x^{\mathrm{SCAN}} \propto -\lambda^{1/2}$ on the wire (eq A15), a functional-level error that can be established without any self-consistent calculation.

DC-SCAN (eqs A1 and A20) adds the log-density curvature tensor C (eq A16), the density Hessian information also used in Hessian-level meta-GGAs.[5] Two scalars are formed from it (eq A18): a rank descriptor D, which equals 1 on the sheet, 2 on the wire and 3 for the isotropic density, and a curvature scale Z. Polynomial weights $w_j(D)$ (eqs A23 and A24) select a 2D branch F_{2D}, a 1D branch F_{1D} (eq A21) or unmodified SCAN, and a smooth switch $\Omega(s, Z)$ (eq A25) activates the branches only where both the reduced gradient and the normalized curvature are large. We fix the two coefficients A_{2D} and A_{1D} (eq A22) analytically on the Gaussian references: A_{2D} reproduces the exact sheet limit (eqs A28–A29), a quasi-2D target in the spirit of ref 4, and $A_{1D} = 1/3$ reproduces the coefficient $\sqrt{2/\pi}$ of the wire logarithm (eqs A31–A32). Because $\Omega = O(s^8)$, the uniform-gas and slowly varying limits of SCAN are untouched, and for $\lambda = 1$ ($D = 3$) DC-SCAN equals SCAN. A curvature-free variant, DC-SCAN$_{\mathrm{min}}$, replaces Ω by Ω_{min} (eq A37). Appendix B lists which exact constraints are retained.

3 Computational Details

Fixed-density quadrature. In cylindrical coordinates the exchange integral reduces to the two-dimensional integral eq A33, in which s, D and Z are known in closed form. We evaluate it with 400 Gauss–Legendre nodes per variable in $\ln u$ and $\ln t$, $u \in [e^{-32}, 26]$, $t \in [e^{-32}, 8]$. Raising the number of nodes to 560 changes DC-SCAN by at most 4.6×10^{-8} Ha on the wire and 7.6×10^{-7} Ha on the sheet (SI, Section S2).

Evaluation inside aniGTO. We take the wire and sheet orbitals as the s functions of the aniGTO trap basis with $\boldsymbol{\omega} = (\lambda^2, \lambda^2, 1)$ and $(1, 1, \lambda^2)$. The exact exchange is the aniGTO repulsion integral $-(\phi\phi|\phi\phi)$; we take the density, gradient, Hessian and τ from analytic

derivatives of the aniGTO basis functions; LDA, PBE, SCAN and r^{2}SCAN exchange come from libxc through PySCF; and D, Z, s and α are formed on the grid from the aniGTO Hessian rather than from the closed form $C = A$ (SI, Section S3). We derive the aniGTO integrals and density derivatives in SI Section S1, and we produce every table and figure with one of eight standalone, single-file Python scripts (SI Section S8).

Logarithmic-variable repulsion integral. aniGTO evaluates the Boys-type representation of the repulsion integral in $t = u/\sqrt{1+u^2} \in [0,1]$. For an exponent ratio ρ_a the integrand is concentrated at $u^2 \sim \rho_a$, i.e. at $1 - t \approx 1/(2\rho_a)$. This region is under-resolved for $\rho_a \geq 10^8$ and lost to round-off beyond $\rho_a \sim 10^{10}$, where $1 - t^2$ rounds to zero. Because $dt\,(1-t^2)^{-3/2} = du$, the same integral reads

$$\int_0^1 I(t)\,dt = \int_{-\infty}^{\infty} e^s \prod_{k=x,y,z} I_k\big(e^{2s}\big)\,ds, \qquad s = \ln u, \tag{1}$$

with the unchanged one-dimensional aniGTO factors I_k. In s the stiff region is an ordinary feature of width $O(1)$, integrated with unit breakpoints over $[\ln\sqrt{a_{\min}} - 20, \ln\sqrt{a_{\max}} + 20]$, plus the analytic contribution of $[0, u_0]$.

Exact $N = 2$ strictly-correlated limit for axisymmetric densities. In cylindrical coordinates the cost $|\mathbf{r}_1 - \mathbf{r}_2|^{-1}$ is bounded below by

$$c(R_1, z_1; R_2, z_2) = \big[(R_1 + R_2)^2 + (z_1 - z_2)^2\big]^{-1/2}, \tag{2}$$

with equality when the partner sits at the opposite azimuth. The three-dimensional SCE problem for an axisymmetric density is therefore exactly a two-dimensional optimal-transport problem in (R, z) with the cost of eq 2. We solve it by stabilized Sinkhorn iterations with rigorous discrete upper (rounded plan) and lower (Kantorovich dual) bounds on grids of spacing h, and extrapolate the two finest grids $h > h'$ to $h \to 0$ with the h^2 Richardson

formula

$$V_{ee}^{\mathrm{SCE}}(0) = V_{ee}^{\mathrm{SCE}}(h') + \frac{V_{ee}^{\mathrm{SCE}}(h') - V_{ee}^{\mathrm{SCE}}(h)}{(h/h')^2 - 1}. \tag{3}$$

For a spherical density $\rho(r)$ with $N = 2$ the exact solution is known in closed form: with $N_e(r) = \int_0^r 4\pi s^2 \rho(s)\, ds$ the co-motion function maps r to the opposite direction at $f(r) = N_e^{-1}\big(2 - N_e(r)\big)$, and

$$V_{ee}^{\mathrm{SCE}} = \int_{N_e(r)<1} \frac{4\pi r^2 \rho(r)}{r + f(r)}\, dr, \tag{4}$$

which provides the exact reference for isotropic Gaussian densities.

4 Results and Discussion

4.1 Quasi-1D Gaussian wires

Table 1: Exchange energies of the two-electron Gaussian wire (eq A8; fixed density, hartree). Exact: closed form (eq A11), confirmed by the aniGTO integral $-(\phi\phi|\phi\phi)$; LDA, SCAN and DC-SCAN (eq A20): 400×400-point logarithmic Gauss–Legendre quadrature (eq A33), converged to 5×10^{-8} Ha. DC-SCAN$_{\mathrm{min}}$ uses the curvature-free switch (eq A37). aniGTO columns: the wire is the aniGTO trap basis with $\boldsymbol{\omega} = (\lambda^2, \lambda^2, 1)$; exact exchange from the aniGTO repulsion integral (default quadrature), and SCAN (libxc), DC-SCAN and DC-PBE (eq A20 on a PBE base) from the density, gradient, Hessian and τ of the aniGTO basis functions on a cylindrical logarithmic grid. Block (a) lists all computed values, block (b) their errors against the exact value of eq A11, and block (c) the implementation checks (aniGTO minus the reference for the same quantity).

(a) Computed exchange energies (Ha)

	exact			reference quadrature				aniGTO basis (eq S3), fields (eq S17), grid (eq S18), libxc					
λ	eq A11	aniGTO, eq S13	Kantorovich, eq S20	LDA	SCAN	DC-SCAN	DC-SCAN$_{\mathrm{min}}$	SCAN	DC-SCAN	PBE	DC-PBE	r^{2}SCAN	DC-r^{2}SCAN
1	-0.797885	-0.797885	-0.797885	-0.681986	-0.795058	-0.795058	-0.795058	-0.795058	-0.795058	-0.763859	-0.763859	-0.795058	-0.795058
2	-1.213337	-1.213337	-1.213337	-1.082585	-1.260632	-1.060135	-1.014727	-1.260632	-1.060135	-1.225359	-0.983006	-1.260632	-1.060135
10	-2.400278	-2.400278	-2.400278	-3.165498	-3.649731	-2.126863	-2.082682	-3.649731	-2.126863	-3.865468	-1.976722	-3.649731	-2.126863
100	-4.227637	-4.227637	-4.227637	-14.692939	-16.362299	-4.193878	-4.185478	-16.362299	-4.193878	-20.817577	-3.992240	-16.362299	-4.193878
1,000	-6.064646	-6.064646	-6.064646	-68.198582	-70.762964	-6.170789	-6.170167	-70.762964	-6.170789	-109.470149	-5.965216	-70.762964	-6.170789
10,000	-7.901840	-7.901907	-7.901840	-316.549776	-292.064785	-7.990438	-7.990407	-292.064785	-7.990438	-545.596537	-7.788276	-292.064785	-7.990438

(b) Error against the exact value of eq A11, $E_x - E_x^{\mathrm{exact}}$ (Ha)

λ	eq A11	aniGTO, eq S13	Kantorovich, eq S20	LDA	SCAN	DC-SCAN	DC-SCAN$_{\mathrm{min}}$	SCAN	DC-SCAN	PBE	DC-PBE	r^{2}SCAN	DC-r^{2}SCAN
1	0	+0.000000	-0.000000	+0.115899	+0.002827	+0.002827	+0.002827	+0.002827	+0.002827	+0.034026	+0.034026	+0.002827	+0.002827
2	0	+0.000000	+0.000000	+0.130752	-0.047295	+0.153201	+0.198609	-0.047295	+0.153201	-0.012022	+0.230330	-0.047295	+0.153201
10	0	+0.000000	+0.000000	-0.765220	-1.249453	+0.273415	+0.317596	-1.249453	+0.273415	-1.465190	+0.423556	-1.249453	+0.273415
100	0	-0.000000	+0.000000	-10.465302	-12.134661	+0.033759	+0.042159	-12.134661	+0.033759	-16.589940	+0.235397	-12.134661	+0.033759
1,000	0	-0.000000	+0.000000	-62.133936	-64.698319	-0.106143	-0.105521	-64.698319	-0.106143	-103.405503	+0.099429	-64.698319	-0.106143
10,000	0	-0.000067	+0.000000	-308.647936	-284.162945	-0.088598	-0.088567	-284.162945	-0.088598	-537.694698	+0.113564	-284.162945	-0.088598

(c) Implementation checks: aniGTO minus reference for the same quantity (Ha)

λ	exact: aniGTO−eq A11	Kant.−eq A11	LDA	SCAN	DC-SCAN	(aniGTO+libxc − reference quadrature)	
1	+0.0e+00	-1.1e-16	+1.7e-14	+2.0e-14	+2.0e-14		
2	+1.1e-15	+0.0e+00	+2.8e-14	+3.3e-14	+4.6e-08		
10	+1.3e-15	+4.4e-16	+8.1e-14	+9.4e-14	-3.2e-08		
100	-3.2e-14	+8.9e-16	+3.7e-13	+4.1e-13	-2.1e-11		
1,000	-3.0e-12	+2.7e-15	+1.7e-12	+1.8e-12	+1.0e-13		
10,000	-6.7e-05	+8.9e-16	+8.1e-12	+7.7e-12	+1.9e-13		

Table 1 shows the consequence of the $\alpha = 0$ degeneracy. Already at $\lambda = 10$ SCAN overbinds the exchange energy by 1.25 Ha, and at $\lambda = 10^4$ it reaches -292 Ha against the exact -7.90 Ha, comparable to the LDA value of -317 Ha; the local exponent $d\ln|E_x|/d\ln\lambda$ falls from 0.64 to 0.59 between $\lambda = 10^2$ and 10^5, toward the predicted 1/2 (SI, Table S2). DC-SCAN removes the power-law divergence: its error is 0.27 Ha at $\lambda = 10$, 0.03 Ha at $\lambda = 100$ and -0.09 to -0.11 Ha for $\lambda = 10^3$–10^4. These values are pre-asymptotic: continuing the same quadrature to $\lambda = 10^{12}$, the slope $dE_x/d\ln\lambda$ converges to the exact $-\sqrt{2/\pi} = -0.797885$ (-0.797878 between 10^{10} and 10^{12}), and the error tends to a constant offset of -0.0403 Ha (-0.0465 at 10^6, -0.0408 at 10^8, -0.0403 at 10^{10} and 10^{12}; SI, Table S3). The leading logarithm is therefore captured, and only a finite-width constant remains; the curvature-free switch is 0.01–0.05 Ha worse for $\lambda = 2$–100 and within 1 mHa beyond.

The aniGTO repulsion integral reproduces the closed-form exact exchange to 3×10^{-12} Ha up to $\lambda = 10^3$ (exponent ratio 10^6); at $\lambda = 10^4$ (ratio 10^8) the default adaptive quadrature of the Boys-type integrand is accurate to 7×10^{-5} Ha and a finer partition of the same integrand restores 8×10^{-11} Ha (SI, Table S1). Beyond $\lambda = 10^4$ the default quadrature fails by several hartree (Table 4), which motivates the logarithmic-variable quadrature of eq 1. With the density, its gradient, Hessian and τ taken from the aniGTO basis functions and SCAN exchange taken from libxc, the SCAN and DC-SCAN columns of the aniGTO block of Table 1 reproduce the reference values to all six printed decimals; the Exact (ERI) column uses the default quadrature, so its $\lambda = 10^4$ entry is off by 6.7×10^{-5} Ha, and the logarithmic-variable quadrature restores -7.901840 (Table 4). This also shows that libxc SCAN and r^{2}SCAN coincide with the closed-form $\alpha = 0$ branch. The transverse-averaged (quasi-1D) Kantorovich route with the transverse-averaged interaction w_b and $b = 1/(\sqrt{2}\lambda)$ is exact for this product density and agrees to 10^{-15} Ha.

The anisotropic Gauss–Hermite grid of the aniGTO DFT module converges SCAN at every λ, but not DC-SCAN for $\lambda \geq 10^3$ (-4.0, -5.8 and -7.9 Ha at $\lambda = 10^4$ for 80, 160 and 240 nodes per axis). The wire logarithm comes from $\lambda^{-2/3} \ll \lambda^2\rho^2 \ll 1$, a region spanning

many decades around the axis that only a logarithmic radial grid resolves (SI, Table S4). On a PBE base the same crossover (DC-PBE) also recovers the logarithm, whereas PBE alone overbinds by 538 Ha at $\lambda = 10^4$. Refitting only the two activation scales of Ω on the wire lowers the DC-SCAN RMS error from 0.154 to 0.027 Ha. We find that this leaves the exact constraints intact and lowers the RMS error on the sheet, which was not fitted, from 0.068 to 0.055 Ha (SI, Table S8). We note that this refit uses the same six wire points it is scored on, so the sheet is its only out-of-sample test.

4.2 Quasi-2D Gaussian sheets

Table 2: Exchange energies of the two-electron Gaussian sheet (eq A7; fixed density, hartree). Exact: closed form (eq A10), confirmed by the aniGTO integral $-(\phi\phi|\phi\phi)$; LDA, SCAN and DC-SCAN (eq A20): the quadrature of Table 1, converged to 8×10^{-7} Ha. The $\lambda \to \infty$ limits are $-\sqrt{\pi/2} = -1.253314$ (exact and DC-SCAN) and -11.145 (SCAN).

λ	Exact	LDA	SCAN	DC-SCAN	DC-SCAN−exact	DC-SCAN$_{\text{min}}$−exact
1	-0.797885	-0.681986	-0.795058	-0.795058	+0.002827	+0.002827
2	-0.964802	-0.859248	-1.000228	-0.869377	+0.095425	+0.117452
10	-1.179303	-1.469294	-1.672690	-1.082995	+0.096308	+0.097156
100	-1.245397	-3.165498	-3.179886	-1.184779	+0.060619	+0.060619
1,000	-1.252517	-6.819858	-5.130958	-1.221408	+0.031109	+0.031109
10,000	-1.253234	-14.692939	-7.045312	-1.238506	+0.014728	+0.014728

We evaluated the same functional, with the same coefficients and no further adjustment, on the Gaussian sheet family, whose orbital is the aniGTO trap function with $\boldsymbol{\omega} = (1, 1, \lambda^2)$ (Table 2). Here the exact exchange stays finite, $E_x^{\text{exact}} \to -\sqrt{\pi/2} + \sqrt{2/\pi}\,\lambda^{-1}$, and so does SCAN: on the $\alpha = 0$ branch its large-s tail $1.174 \cdot 4.9479\, s^{-1/2}$ has the form of F_{2D}, so the integral that fixes A_{2D} (eq A28) also gives $E_x^{\text{SCAN}} \to -1.174 \cdot 4.9479\,\sqrt{\pi/2}/A_{2D} = -11.145$ Ha, 8.9 times the exact limit. This limit is approached slowly: at $\lambda = 10^4$ SCAN is -7.05 Ha, an error of -5.79 Ha, while LDA diverges as $\lambda^{1/3}$ (-14.7 Ha). DC-SCAN stays within 0.10 Ha of the exact energy at every λ; its error, +0.095, +0.096, +0.061, +0.031 and +0.015 Ha for $\lambda = 2$, 10, 100, 10^3 and 10^4, decays as $\lambda^{-1/3}$ (+0.0032 Ha at $\lambda = 10^6$; SI, Table S6), so the finite sheet limit built into A_{2D} is reached and, unlike on the wire, no finite-width offset survives. The curvature-free switch differs only at $\lambda = 2$ (+0.117 Ha) and

agrees within 1 mHa from $\lambda = 10$. The aniGTO repulsion integral reproduces the closed-form sheet exchange to 10^{-15} Ha up to $\lambda = 10^3$, with the same default-quadrature error at $\lambda = 10^4$ (8×10^{-5} Ha) and the same cure as for the wire (SI, Table S5).

We validated the sheet construction against independent literature results. Applied unchanged to the infinite-barrier slab of refs 2,3, the formula that fixes A_{2D} (eqs A28–A29) gives the coefficient $a_x = 0.521748$ for the exact 2D limit, against 0.5217 derived by Constantin,[2] and the SCAN 2D limit $E_x/N = -1.6706$ Ha at $r_s^{2D} = 4$, against -1.671 Ha (analytic) and -1.655 Ha (numerical) of Kaplan, Wagle and Perdew,[3] with the exact value -0.1501 Ha (Table 3; SI, Table S7). For the Gaussian sheet the aniGTO exact exchange at $\lambda = 10^8$, -1.253314129337 Ha, equals the closed form to 12 digits and gives the same A_{2D} and SCAN/exact ratio as the analytic limit. SCAN thus overbinds the 2D limit by a factor of 11.1 for the slab and 8.9 for the Gaussian sheet: the failure follows from the large-s form of SCAN, $F_x \to h_x^0 a_1 s^{-1/2}$, and not from our choice of model density, whose shape enters only through the coefficient (0.522 for the $\sin^2$ slab, 0.653 for the Gaussian).

Table 3: Validation against independent literature results: the sheet-limit formula that fixes A_{2D} (eqs A28–A29; SI, eq S21) applied unchanged to the infinite-barrier slab of refs 2,3 and to the Gaussian sheet of this work. aniGTO: the same quantity from the aniGTO exact exchange of the Gaussian sheet at $\lambda = 10^8$ (self-repulsion integral, SI eqs S14–S15, in the logarithmic variable of eq 1); "–": not applicable, because the $\sin^2$ slab is not a Gaussian density. Computed values first, then the literature or reference value, its source, and the relative differences at the end.

	this work				difference to reference	
quantity	eq S21	aniGTO	reference	source	eq S21	aniGTO
slab: exact 2D E_x/N (Ha), $r_s^{2D} = 4$	-0.150053	–	-0.1501	ref 3	+0.03%	–
slab: a_x giving the exact 2D limit	0.521748	–	0.5217	ref 2, eq 12	+0.01%	–
slab: SCAN 2D limit E_x/N (Ha), analytic	-1.670599	–	-1.6710	ref 3, App. A	+0.02%	–
slab: SCAN 2D limit E_x/N (Ha), numerical	-1.670599	–	-1.6550	ref 3, numerical	−0.94%	–
slab: SCAN/exact	11.133411	–	11.0260	ref 3	+0.97%	–
Gaussian sheet: exact 2D limit E_x (Ha)	-1.253314	-1.253314	-1.2533	eq A12	+0.00%	+0.00%
Gaussian sheet: A_{2D}	0.653226	0.653226	0.6532	eq A29	+0.00%	−0.00%
Gaussian sheet: SCAN limit (Ha)	-11.145141	–	-11.1451	eq A28, Table 2	+0.00%	–
Gaussian sheet: SCAN/exact	8.892536	8.892536	8.8900	main text	+0.03%	+0.03%

4.3 Exact references versus the proposed numerical methods

Table 4 and Figure 1 compare the two proposed numerical methods of Section 3 with exact references. For the repulsion integral, the default quadrature degrades from 3×10^{-11} mHa at $\lambda = 10^2$ to 0.067 mHa at 10^4, and fails by 5.8–11.3 Ha for $\lambda = 10^5$–10^8, where it returns an almost constant -3.906 Ha (Figure 1a). The logarithmic variable reproduces the closed form to $< 10^{-11}$ mHa, the double-precision limit, at every λ for both the wire and the sheet. It agrees with the default to machine precision on an ordinary anisotropic Hookium (d, g) quartet (exponent ratio 5), at about 1.7 times the cost. For the $N = 2$ SCE problem of isotropic Gaussian densities, where the radial co-motion solution is exact, the grid error of the axisymmetric optimal transport decreases as h^2, and the extrapolated V_{ee}^{SCE} agrees with the exact value to 0.030 and 0.009 mHa (Figure 1c). The same solver applies unchanged to prolate and oblate traps, where it replaces a radialized (spherically averaged) W_∞ by the exact anisotropic value.

Table 4: Exact references versus the proposed methods (hartree). (a,b) Self-repulsion $E_x = -(\phi\phi|\phi\phi)$ of the two-electron Gaussian wire and sheet. *Exact*: closed forms eq A11 (wire) and eq A10 (sheet). *Baseline, "aniGTO default"*: the repulsion integral as computed by the aniGTO package unchanged, i.e. its generalized-Boys integrand of eq S13 integrated over $t \in [0, 1]$ with adaptive quadrature (`scipy.integrate.quad`). *Proposed*: also aniGTO – the same one-dimensional aniGTO kernels I_k (eq S12), integrated in the logarithmic variable $s = \ln u$ of eq 1; only the integration variable changes. (c) $N = 2$ SCE interaction energy of isotropic Gaussian densities $2(\omega/\pi)^{3/2} e^{-\omega r^2}$. *Exact*: radial co-motion solution, eq 4. *Baseline*: axisymmetric optimal transport with the cost of eq 2 on the finest grid. *Proposed*: the same transport extrapolated to $h \to 0$, eq 3. Errors $|E - E^{\text{exact}}|$ in mHa, $< 10^{-11}$ mHa being the double-precision limit; time is the wall time of one self-repulsion integral in ms.

			baseline	proposed	$\lvert E - E^{\text{exact}} \rvert$ (mHa)		time per integral (ms)	
quantity	parameter	exact	(see block)	(see block)	of baseline	of proposed	of baseline	of proposed
(a) wire, $E_x = -(\phi\phi\|\phi\phi)$. Exact: eq A11; baseline: aniGTO default (eq S13, adaptive in t); proposed: aniGTO kernels in $s = \ln u$ (eq 1)								
wire	$\lambda = 10^1$	-2.400277840	-2.400277840	-2.400277840	$< 10^{-11}$	$< 10^{-11}$	20	73
wire	$\lambda = 10^2$	-4.227637064	-4.227637064	-4.227637064	3.2×10^{-11}	$< 10^{-11}$	40	79
wire	$\lambda = 10^3$	-6.064645553	-6.064645553	-6.064645553	3.0×10^{-9}	$< 10^{-11}$	55	80
wire	$\lambda = 10^4$	-7.901839854	-7.901906791	-7.901839854	0.067	$< 10^{-11}$	68	81
wire	$\lambda = 10^5$	-9.739036912	-3.906470773	-9.739036912	5833	$< 10^{-11}$	5	88
wire	$\lambda = 10^6$	-11.576234008	-3.906470882	-11.576234008	7670	$< 10^{-11}$	5	93
wire	$\lambda = 10^7$	-13.413431103	-3.906470883	-13.413431103	9507	$< 10^{-11}$	5	94
wire	$\lambda = 10^8$	-15.250628199	-3.906470883	-15.250628199	11344	$< 10^{-11}$	5	99
(b) sheet, $E_x = -(\phi\phi\|\phi\phi)$. Exact: eq A10; baseline and proposed as in (a)								
sheet	$\lambda = 10^1$	-1.179303431	-1.179303431	-1.179303431	$< 10^{-11}$	$< 10^{-11}$	54	70
sheet	$\lambda = 10^2$	-1.245397430	-1.245397430	-1.245397430	$< 10^{-11}$	$< 10^{-11}$	41	74
sheet	$\lambda = 10^3$	-1.252516879	-1.252516879	-1.252516879	$< 10^{-11}$	$< 10^{-11}$	62	79
sheet	$\lambda = 10^4$	-1.253234355	-1.253314143	-1.253234355	0.080	$< 10^{-11}$	61	83
sheet	$\lambda = 10^5$	-1.253306159	-1.253314137	-1.253306159	8.0×10^{-3}	$< 10^{-11}$	30	88
sheet	$\lambda = 10^6$	-1.253313339	-1.253314137	-1.253313339	8.0×10^{-4}	$< 10^{-11}$	21	94
sheet	$\lambda = 10^7$	-1.253314058	-1.253314137	-1.253314058	8.0×10^{-5}	$< 10^{-11}$	18	92
sheet	$\lambda = 10^8$	-1.253314129	-1.253314137	-1.253314129	7.9×10^{-6}	$< 10^{-11}$	17	97
(c) $N = 2$ SCE, V_{ee}^{SCE}. Exact: eq 4; baseline: optimal transport (eq 2), finest grid; proposed: $h \to 0$, eq 3								
Gaussian	$\omega = 1$	0.444142598	0.444522674	0.444172297	0.380	0.030	–	–
Gaussian	$\omega = 0.1$	0.140450222	0.140570412	0.140459613	0.120	9.4×10^{-3}	–	–

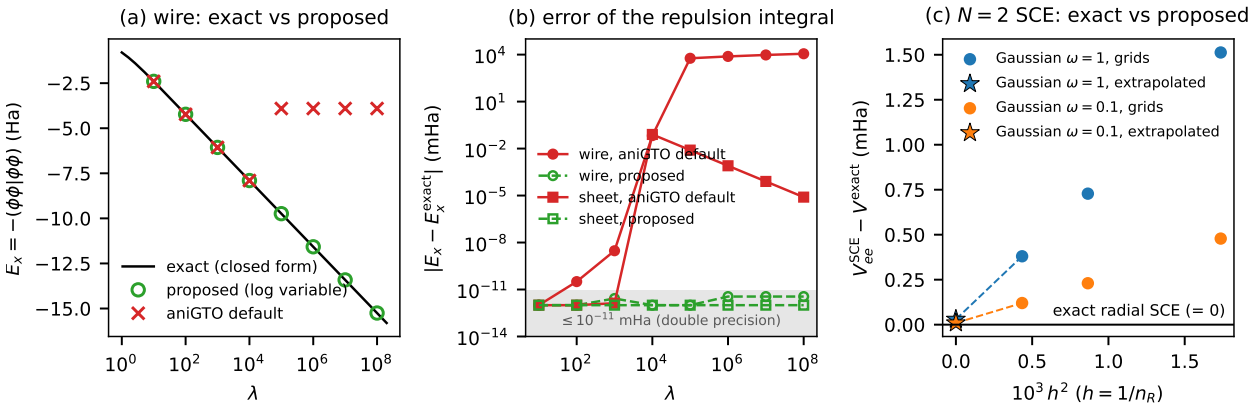


Figure 1: Matching of exact references and proposed methods. (a) Wire self-repulsion versus λ: closed form (line), logarithmic-variable quadrature (circles) and aniGTO default quadrature (crosses), which collapses to -3.906 Ha for $\lambda \geq 10^5$. (b) Absolute errors (mHa) for the wire and the sheet; the shaded band is the double-precision limit. (c) $N = 2$ SCE energy of isotropic Gaussians: axisymmetric optimal-transport values on three grids (circles) against h^2, and the extrapolated values (stars), which coincide with the exact radial solution (zero line) to 0.03 and 0.009 mHa.

4.4 Beyond the semilocal form

We tested two routes beyond the semilocal form on the same wires (SI, Section S6); our analysis shows that range-separated hybrids do not help, because 90–98% of the wire exchange is short-range at the HSE06 screening parameter. First, a quasi-1D LDA built on the Kantorovich kernel w_b, with b taken from the density curvature, reproduces the wire logarithm exactly; its error, +120.2 mHa for every $\lambda \geq 100$, is a λ-independent constant from the axial inhomogeneity, and a one-parameter 1D gradient correction fitted on the Gaussian removes it there (0.03 mHa) but not on other axial profiles. Second, a local hybrid that mixes in exact exchange with the iso-orbital weight $a = \tau_W/\tau$ is exact for $N = 2$ (0.000 mHa at every λ). For $N = 4$ and 6 wires built from the s, p_z and d_{zz} functions of the same aniGTO trap basis, with the exact exchange-energy density from the same anisotropic integrals, its error falls from tens of mHa at $\lambda \leq 100$ to -0.10 and -0.30 mHa (LH-DC-SCAN) at $\lambda = 10^4$, because $a \to 1$ once the transverse kinetic energy dominates. DC-SCAN alone, whose 1D branch was fixed on $N = 2$, misses by -0.44 and -0.93 Ha there.

5 Conclusions

In this study, we have shown that on one-orbital densities the SCAN indicator α vanishes and cannot distinguish confinement geometries. On exactly solvable two-electron Gaussian densities this makes SCAN exchange diverge as $\lambda^{1/2}$ under wire confinement, instead of logarithmically, and converge to a sheet limit 8.9 times too negative; applied to the infinite-barrier slab, the same analysis reproduces the independent results of Constantin and of Kaplan, Wagle and Perdew to 0.02%. We have developed DC-SCAN, in which the log-density curvature supplies the missing geometric information. With two coefficients fixed analytically on the Gaussian references, our DC-SCAN candidate captures the exact wire logarithm up to a finite-width offset of -0.040 Ha and reaches the exact sheet limit, while leaving the uniform-gas and slowly varying limits of SCAN unchanged. Our evaluation of the same functional inside aniGTO exposed two numerical requirements: a logarithmic radial grid around the confinement axis for Hessian-dependent functionals, and a logarithmic-variable repulsion integral for exponent ratios beyond 10^8. We stress that these are fixed-density, exchange-only results on one Gaussian profile per family. Other confinement profiles, three-dimensional norms (H, He, Ne, jellium), smooth regularization of D and $\ln n$, the self-consistent potential (Appendix A.9) and correlation remain to be tested, and we will address them before DC-SCAN can be considered a transferable functional.

Appendix A: Theory and Equations of DC-SCAN

This appendix collects all definitions and derivations of the DC-SCAN candidate; eqs A1–A44 correspond one-to-one, in order, to Eqs. (1)–(44) of the DC-SCAN technical note (`reference/DC_SCAN_detailed_theory_results.pdf`). Throughout, the Coulomb interaction is the three-dimensional $1/r_{12}$ interaction and all coordinates are in bohr.

A.1 Main working equations

$$E_x^{\text{DC-SCAN}}[n] = -C_x \int d^3r\, n^{4/3}(\mathbf{r})\, F_x^{\text{DC-SCAN}}(s,\alpha,D,Z), \tag{A1}$$

$$F_x^{\text{DC-SCAN}} = \big[1-\Omega(w_{2D}+w_{1D})\big] F_x^{\text{SCAN}}(s,\alpha) + \Omega\big[w_{2D}F_{2D}(s) + w_{1D}F_{1D}(s)\big], \tag{A2}$$

where $C_x = \frac{3}{4}(3/\pi)^{1/3}$, $F_{2D}(s) = 0.65322589699\,(1+s^2)^{-1/4}$, $F_{1D}(s) = \{3(1+s^2)\}^{-1}$, and D, Z, w_j and Ω are defined below. These numerical coefficients follow from the chosen two-electron Gaussian densities; they are not universal constants of an exact exchange functional.

A.2 SCAN exchange

Standard SCAN exchange[1] is

$$E_x^{\text{SCAN}}[n] = -C_x \int n^{4/3} F_x^{\text{SCAN}}(s,\alpha)\, d^3r, \tag{A3}$$

$$s = \frac{|\nabla n|}{2k_F n}, \qquad k_F = (3\pi^2 n)^{1/3}, \tag{A4}$$

$$\alpha = \frac{\tau - \tau_W}{\tau_{\text{unif}}}, \qquad \tau_W = \frac{|\nabla n|^2}{8n}, \qquad \tau_{\text{unif}} = \frac{3}{10}k_F^2 n. \tag{A5}$$

For a spin-unpolarized two-electron state with a single doubly occupied real spatial orbital, $\tau = \tau_W$ and $\alpha = 0$, irrespective of whether the density is spherical, sheet-like or wire-like. On this branch

$$F_x^{\text{SCAN}}(s,0) = 1.174\big[1-\exp(-4.9479/\sqrt{s})\big] \underset{s\to\infty}{\sim} 5.8088346\, s^{-1/2}. \tag{A6}$$

A.3 Exact two-electron reference densities

The normalized reference families are

$$n_\lambda^{\text{sheet}}(x,y,z) = \frac{2\lambda}{\pi^{3/2}} \exp[-(x^2+y^2+\lambda^2 z^2)], \tag{A7}$$

$$n_\lambda^{\text{wire}}(x,y,z) = \frac{2\lambda^2}{\pi^{3/2}} \exp[-(\lambda^2(x^2+y^2)+z^2)], \tag{A8}$$

with $\lambda \geq 1$. They integrate to $N = 2$ and are each generated by one real doubly occupied orbital, so $\alpha = 0$ wherever $n > 0$. The exchange-only exact reference is

$$E_x^{\text{exact}}[n] = -\frac{1}{2} E_H[n] = -\frac{1}{4} \iint \frac{n(\mathbf{r})\, n(\mathbf{r}')}{|\mathbf{r}-\mathbf{r}'|}\, d^3r\, d^3r'. \tag{A9}$$

Gaussian Coulomb integration (Appendix A.12) gives, for $\lambda > 1$,

$$E_x^{\text{sheet,exact}}(\lambda) = -\sqrt{\frac{2}{\pi}} \frac{\lambda}{\sqrt{\lambda^2-1}} \arctan\sqrt{\lambda^2-1}, \tag{A10}$$

$$E_x^{\text{wire,exact}}(\lambda) = -\sqrt{\frac{2}{\pi}} \frac{\lambda}{\sqrt{\lambda^2-1}} \operatorname{arcosh}\lambda. \tag{A11}$$

Both approach $-\sqrt{2/\pi}$ Ha as $\lambda \to 1$. Their strongly confined limits differ:

$$E_x^{\text{sheet,exact}}(\lambda) \longrightarrow -\sqrt{\frac{\pi}{2}} \qquad (\lambda \to \infty), \tag{A12}$$

$$E_x^{\text{wire,exact}}(\lambda) = -\sqrt{\frac{2}{\pi}} \ln(2\lambda) + O\left(\frac{\ln\lambda}{\lambda^2}\right). \tag{A13}$$

Sheet compression gives a finite exchange energy with the 3D $1/r$ interaction; the zero-width wire retains a logarithmic Coulomb singularity.

A.4 Why the SCAN branch fails on the wire

At generic scaled positions the reduced gradient behaves as

$$s_{\text{sheet}} \sim \lambda^{2/3}, \qquad s_{\text{wire}} \sim \lambda^{1/3}, \tag{A14}$$

and the exchange volume factors scale as $n^{4/3}d^3r \sim \lambda^{1/3}$ and $\lambda^{2/3}$, respectively. Inserting $F_x(s,0) \sim s^{-1/2}$ gives a finite sheet limit but

$$E_x^{\text{SCAN}}[n_\lambda^{\text{wire}}] \sim -\text{const} \cdot \lambda^{1/2}, \tag{A15}$$

rather than the logarithm of eq A13.

A.5 Log-density curvature and geometry descriptors

The symmetric log-density curvature tensor is

$$C_{ij} = -\frac{1}{2}\partial_i\partial_j \ln n = -\frac{n_{ij}}{2n} + \frac{n_i n_j}{2n^2}, \qquad n_i = \partial_i n, \quad n_{ij} = \partial_i\partial_j n. \tag{A16}$$

For the two Gaussian families it is position independent,

$$C^{\text{sheet}} = \text{diag}(1, 1, \lambda^2), \qquad C^{\text{wire}} = \text{diag}(\lambda^2, \lambda^2, 1). \tag{A17}$$

The directional effective-rank descriptor and the curvature scale are

$$D = \text{clip}_{[1,3]}\left(\frac{|\nabla n|^2 \,\text{Tr}(C^2)}{(\nabla n)^T C^2 (\nabla n)}\right), \qquad Z = \frac{\sqrt{\text{Tr}(C^2)}}{k_F^2}, \tag{A18}$$

with $D = 3$ where the denominator vanishes. At generic points of the Gaussian families

$$D \to 1 \text{ (sheet)}, \qquad D \to 2 \text{ (wire)}, \qquad D = 3 \text{ (isotropic)}. \tag{A19}$$

Under uniform coordinate scaling $n_\gamma(\mathbf{r}) = \gamma^3 n(\gamma\mathbf{r})$, $C \mapsto \gamma^2 C$ and $k_F^2 \mapsto \gamma^2 k_F^2$ at scaled positions, so D and Z are dimensionless scale-invariant fields; both are rotational scalars. For the hydrogenic density $n \propto e^{-2\zeta r}$, C has zero radial curvature and ∇n is radial, and the convention gives $D = 3$.

A.6 DC-SCAN enhancement factor

$$F_x^{\text{DC-SCAN}}(s,\alpha,D,Z) = \big[1 - \Omega(s,Z)(w_{2D} + w_{1D})\big] F_x^{\text{SCAN}}(s,\alpha) + \Omega(s,Z)\big[w_{2D}F_{2D}(s) + w_{1D}F_{1D}(s)\big], \tag{A20}$$

$$F_{2D}(s) = \frac{A_{2D}}{(1+s^2)^{1/4}}, \qquad F_{1D}(s) = \frac{A_{1D}}{1+s^2}, \tag{A21}$$

$$A_{2D} = 0.6532258969913366, \qquad A_{1D} = \frac{1}{3}. \tag{A22}$$

The geometry weights are built from the polynomials

$$L_{2D}(D) = \tfrac{1}{2}(D-2)(D-3), \qquad L_{1D}(D) = (D-1)(3-D), \tag{A23}$$

$$L_{3D}(D) = \tfrac{1}{2}(D-1)(D-2), \qquad w_j(D) = \frac{L_j(D)^2}{L_{2D}^2 + L_{1D}^2 + L_{3D}^2}. \tag{A24}$$

They satisfy $0 \le w_j \le 1$ and $w_{2D} + w_{1D} + w_{3D} = 1$ and select the 2D, 1D and unmodified SCAN branches at $D = 1$, 2 and 3. The activation switch is

$$\Omega(s,Z) = \frac{Z^3}{1+Z^3}\,\frac{s^8}{1+s^8}; \tag{A25}$$

the exponents 3 and 8 are design choices, and no molecular or solid-state data are fitted. An equivalent form displays the convex mixture,

$$F_x^{\text{DC-SCAN}} = (1-\Omega)F_x^{\text{SCAN}} + \Omega\big[w_{2D}F_{2D} + w_{1D}F_{1D} + w_{3D}F_x^{\text{SCAN}}\big]. \tag{A26}$$

If the base SCAN enhancement obeys a pointwise upper bound, this form preserves the bound provided each added branch obeys it; both branches have maxima below 1.174.

A.7 Analytic choice of the coefficients

Sheet. Let $c = 2/\pi^{3/2}$ and $K = (3\pi^2)^{1/3}$. With $u = x^2 + y^2$ and $t = \lambda z$, the sheet has $n = \lambda c \exp(-u - t^2)$ and

$$s = \frac{\sqrt{u + \lambda^2 t^2}}{K n^{1/3}} \sim \frac{\lambda^{2/3}|t|}{K[c\exp(-u - t^2)]^{1/3}} \qquad (t \neq 0). \tag{A27}$$

For $F_{2D} \sim A_{2D} s^{-1/2}$ the integrated exchange tends to $-A_{2D} M_{2D}$, with

$$M_{2D} = C_x \sqrt{K}\, c^{3/2} \left(\frac{2\pi}{3}\right) \left(\frac{3}{2}\right)^{-1/4} \Gamma\left(\frac{1}{4}\right) = 1.918653475142493. \tag{A28}$$

Matching eq A12 fixes

$$A_{2D} = \frac{\sqrt{\pi/2}}{M_{2D}} = 0.6532258969913366. \tag{A29}$$

The 2D Gaussian reference limit, including its coefficient, is thus imposed analytically; it is not an exact result for an arbitrary 2D electron gas.

Wire. With $u = \lambda^2(x^2 + y^2)$ and $t = z$, $n = \lambda^2 c \exp(-u - t^2)$ and

$$s \sim \frac{\lambda^{1/3}\sqrt{u}}{K[c\exp(-u - t^2)]^{1/3}} \qquad (u > 0). \tag{A30}$$

When $F_{1D} \sim A_{1D}/s^2$, the radial integral has a logarithmic region $\lambda^{-2/3} \ll u \ll 1$, in which du/u supplies $\frac{2}{3}\ln\lambda$, and

$$E_x^{1D} \sim -\underbrace{\left[\frac{2\pi}{3} C_x K^2 c^2 \sqrt{\frac{\pi}{2}}\right]}_{3\sqrt{2/\pi}} A_{1D} \ln\lambda + O(1). \tag{A31}$$

Matching the exact coefficient $\sqrt{2/\pi}$ of eq A13 gives

$$A_{1D} = \frac{1}{3}. \tag{A32}$$

This fixes the coefficient of the logarithm, not the finite-width energy or its $O(1)$ constant.

A.8 Fixed-density quadrature

In cylindrical coordinates and the transformed variables $u, t \geq 0$ above,

$$E_x = -2\pi C_x c^{4/3} \lambda^{k/3} \int_0^\infty du \int_0^\infty dt \; \exp\left[-\tfrac{4}{3}(u+t^2)\right] F_x(s, \alpha = 0, D, Z), \tag{A33}$$

with $k = 1$ for the sheet and $k = 2$ for the wire.

A.9 Self-consistent functional derivative

The energy density is $e_x = -C_x n^{4/3} F_x^{\text{DC-SCAN}}(n, \nabla n, \nabla\nabla n, \tau)$. Its density-dependent Euler derivative is

$$v_x^{\text{dens}} = \frac{\partial e_x}{\partial n} - \partial_i \left(\frac{\partial e_x}{\partial n_i} \right) + \partial_i \partial_j \left(\frac{\partial e_x}{\partial n_{ij}} \right). \tag{A34}$$

The τ dependence generates the usual generalized Kohn–Sham operator, which acts on an orbital as

$$\tau_\sigma = \frac{1}{2} \sum_{i \in \sigma} f_i |\nabla \phi_i|^2, \qquad \hat{V}_{\tau,\sigma} \phi_{i\sigma} = -\frac{1}{2} \nabla \cdot \left[\frac{\partial e_x}{\partial \tau_\sigma} \nabla \phi_{i\sigma} \right]. \tag{A35}$$

Exact exchange spin scaling defines the spin-dependent functional,

$$E_x[n_\uparrow, n_\downarrow] = \tfrac{1}{2} E_x^{\text{DC-SCAN}}[2n_\uparrow, 2\tau_\uparrow] + \tfrac{1}{2} E_x^{\text{DC-SCAN}}[2n_\downarrow, 2\tau_\downarrow], \tag{A36}$$

where the second argument is the kinetic-energy-density input of each spin-channel evaluation. The Hessian dependence introduces up to fourth-order spatial derivatives into the local potential.

A.10 Minimal-information variant

To test whether the curvature magnitude Z is needed, the same D and weights are kept and $\Omega(s, Z)$ is replaced by

$$\Omega_{\min}(s) = \frac{s^8}{1+s^8}. \tag{A37}$$

A.11 Explicit SCAN exchange ingredients

$$F_x^{\mathrm{SCAN}}(s,\alpha) = \left\{h_x^1(s,\alpha) + f_x(\alpha)\left[h_x^0 - h_x^1(s,\alpha)\right]\right\} g_x(s), \tag{A38}$$

$$h_x^0 = 1.174, \qquad g_x(s) = 1 - \exp(-a_1/\sqrt{s}), \tag{A39}$$

$$h_x^1(s,\alpha) = 1 + k_1 - \frac{k_1}{1 + x(s,\alpha)/k_1}, \tag{A40}$$

$$x(s,\alpha) = \mu s^2 \left[1 + \frac{b_4 s^2}{\mu} e^{-|b_4| s^2/\mu}\right] + \left[b_1 s^2 + b_2(1-\alpha) e^{-b_3(1-\alpha)^2}\right]^2, \tag{A41}$$

$$f_x(\alpha) = \begin{cases} \exp[-c_1\alpha/(1-\alpha)], & \alpha < 1, \\ 0, & \alpha = 1, \\ -d_x \exp[c_2/(1-\alpha)], & \alpha > 1, \end{cases} \tag{A42}$$

with $a_1 = 4.9479$, $k_1 = 0.065$, $\mu = 10/81$, $b_1 \simeq 0.1566$, $b_2 \simeq 0.1208$, $b_3 = 0.5$, $b_4 \simeq 0.1218$, $c_1 = 0.667$, $c_2 = 0.8$ and $d_x = 1.24$; $g_x(0) = 1$ by continuity. The Gaussian tests use only $\alpha = 0$, on which eq A6 is exact.

A.12 Gaussian Coulomb formulas

For an anisotropic Gaussian density $n_A(\mathbf{r}) = 2\sqrt{\det A}\,\pi^{-3/2} e^{-\mathbf{r}^T A \mathbf{r}}$ with $A > 0$, write

$$\frac{1}{r_{12}} = \frac{2}{\sqrt{\pi}} \int_0^\infty e^{-t^2 r_{12}^2}\, dt. \tag{A43}$$

The difference of two independently distributed Gaussian positions has covariance A^{-1}, giving

$$E_x[n_A] = -\frac{2}{\sqrt{\pi}} \int_0^\infty \frac{\sqrt{\det A}}{\sqrt{\det(A + 2t^2 I)}}\, dt. \tag{A44}$$

With $A = \mathrm{diag}(1, 1, \lambda^2)$ or $\mathrm{diag}(\lambda^2, \lambda^2, 1)$ elementary integration yields eqs A10 and A11; the $\lambda = 1$ limit follows by continuity.

Appendix B: Constraint Audit

Table B1: Status of the DC-SCAN exchange construction. "Retained" applies to the definitions and assumptions of Appendix A, not to all SCAN exchange–correlation conditions.

Property	Status / reason
Uniform electron gas	Retained: $s = 0$ implies $\Omega = 0$, so $F_x = F_x^{\mathrm{SCAN}} = 1$.
Uniform coordinate scaling	Retained: s, α, D and Z are dimensionless under scaling.
Rotational invariance	Retained: tensor contractions are scalars.
Exchange spin scaling	Retained with eq A36.
Gradient expansion to fourth order	Retained in a smooth slowly varying limit: $\Omega = O(s^8)$.
Pointwise bound $F_x \leq 1.174$	Retained if SCAN obeys it: both branches stay below 1.174 and eq A26 is a convex mixture.
Hydrogenic exponential density	Unmodified through $D = 3$; other atomic norms not established.
Gaussian sheet energy limit	Imposed analytically through A_{2D}; checked numerically.
Gaussian wire leading logarithm	Imposed analytically through A_{1D}; checked numerically.
Exact exchange for arbitrary 2D/1D densities	Not established; coefficients depend on the reference profiles.
Exchange hole, potential regularity, nonuniform scaling, all SCAN constraints	Not established.

A local integral of finitely many density derivatives is additive for disjoint supports, whereas the two-electron exchange $-E_H/2$ contains the Coulomb cross term between separated lobes; exact exchange for all two-electron densities therefore requires nonlocal information. In practice $\ln n$ and C are ill-conditioned in low-density tails and the clipping of D must be replaced by a differentiable regularization before a self-consistent potential can be used.

Supporting Information Available

Numerical protocol and convergence checks; exact-exchange cross-checks for the wire and the sheet; local slopes and the finite-width offset on the wire; aniGTO grid comparison; sheet asymptotics; PBE base and refit of the activation scales; range separation, quasi-1D LDA/GGA and local hybrids; figures of the wire and sheet benchmarks (PDF).

References

(1) Sun, J.; Ruzsinszky, A.; Perdew, J. P. Strongly constrained and appropriately normed semilocal density functional. *Phys. Rev. Lett.* **2015**, *115*, 036402, DOI: 10.1103/PhysRevLett.115.036402.

(2) Constantin, L. A. Dimensional crossover of the exchange-correlation energy at the semilocal level. *Phys. Rev. B* **2008**, *78*, 155106, DOI: 10.1103/PhysRevB.78.155106.

(3) Kaplan, A. D.; Wagle, K.; Perdew, J. P. Collapse of the electron gas from three to two dimensions in Kohn-Sham density functional theory. *Phys. Rev. B* **2018**, *98*, 085147, DOI: 10.1103/PhysRevB.98.085147.

(4) Horowitz, C. M.; Proetto, C. R.; Pitarke, J. M. Construction of a semilocal exchange density functional from a three-dimensional electron gas collapsing to two dimensions. *Phys. Rev. B* **2023**, *108*, 115119, DOI: 10.1103/PhysRevB.108.115119.

(5) Dabbaghi, P.; García Lastra, J. M.; de Silva, P. Self-consistent Hessian-level meta-generalized gradient approximation. *Phys. Rev. B* **2026**, DOI: 10.1103/x4qn-mbr1.

(6) Chirchir, G.; Jana, S.; 'Smiga, S. An Anisotropic Gaussian Integral Package for Quasi-Dimensional Quantum-Chemical Simulations. 2026; Manuscript in preparation.